

I tried to split the csv (Survey Data) file using Python for upload to Google Looker, it did work but it was still over 100MB

```
C: > Users > Brendan > OneDrive > Documents > IBM Data Analyst > CSV split.py > ...
1  import pandas as pd
2  import os
3
4  # Path to your big csv file
5  input_file = r"C:\Users\Brendan\Downloads\survey-data.csv"
6
7  # Output folder (change if you want a different location)
8  output_folder = r"C:\Users\Brendan\Downloads"
9
10 # Load the csv
11 df = pd.read_csv(input_file)
12
13 # Define chunk size (number of rows per file)
14 chunk_size = 25000 # adjust if needed
15
16 # Loop through and save chunks
17 for i in range(0, len(df), chunk_size):
18     chunk = df.iloc[i:i+chunk_size]
19     output_path = os.path.join(output_folder, f"survey-data-part{i//chunk_size+1}.csv")
20     chunk.to_csv(output_path, index=False)
21     print(f"Saved: {output_path}")
22
```

DASHBOARD

Uploaded Updated file from Github (Alexander Booth, you legend!)

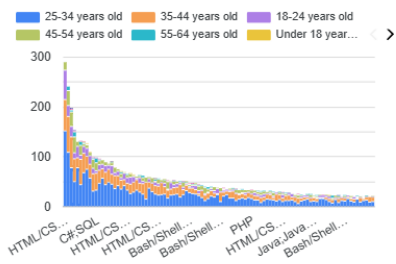
The screenshot shows the Google Data Studio interface. At the top, there's a header "Add data to report" with a back arrow and a refresh icon. Below this, a blue banner indicates "1,95 GB space available out of 2 GB." The main content area is titled "CSV File Upload" with a sub-header "By Google". It explains that users can bring data into Data Studio by uploading CSV files. Below this, there are links for "Learn more" and "Report an issue". The interface is divided into three main sections: "Datasets", "Files", and "Schema". The "Datasets" section shows a search bar and a list of datasets, with "Survey data" selected. The "Files" section shows a search bar and a list of files, with "survey_data_updated" selected. The "Schema" section shows the fields and their order in the data set, including "Responseld", "MainBranch", "Age", "Employment", "RemoteWork", "Check", "CodingActivities", and "EdLevel".

I worked alongside the tutor so the slides should look similar to what he did

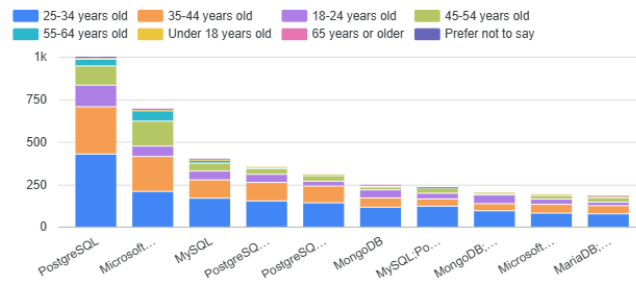
Current Tech Usage

Select date range

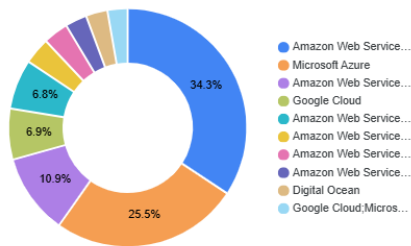
Record Count by LanguageHaveWorkedWith...



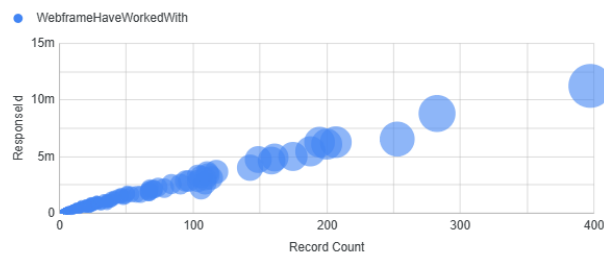
Record Count by DatabaseHaveWorkedWith and Age



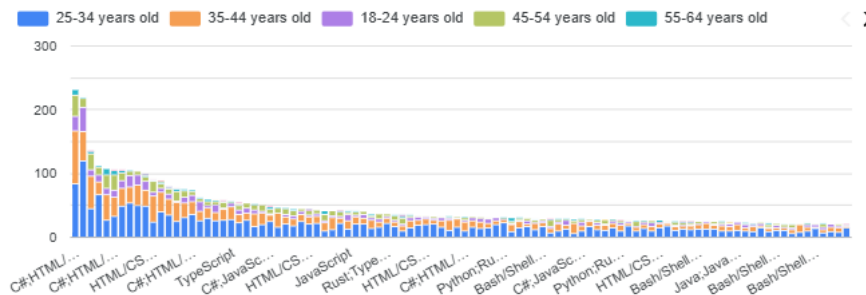
Platform Have Worked With



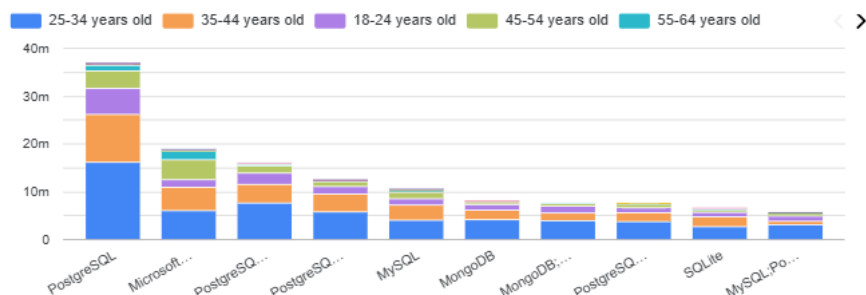
Distribution of Record Count by Responsed, size indicates Record Count



Record Count by LanguageWantToWorkWith and Age

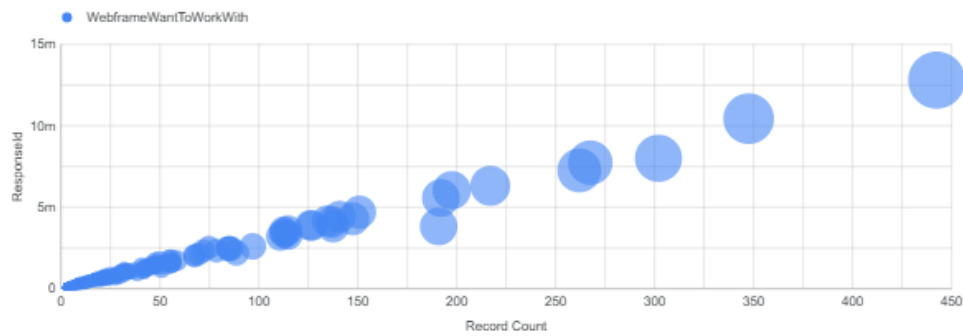


Responsed by DatabaseWantToWorkWith and Age



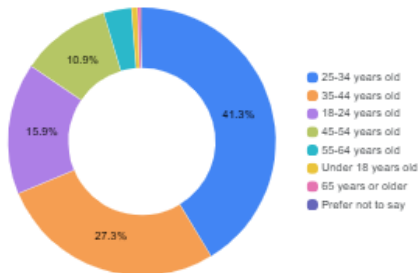
Future Tech Usage

Record Count by PlatformWantToWorkWith



Demographics

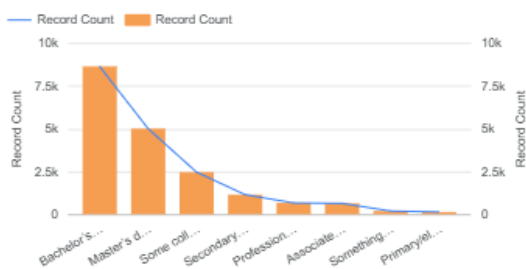
Age by Record Count



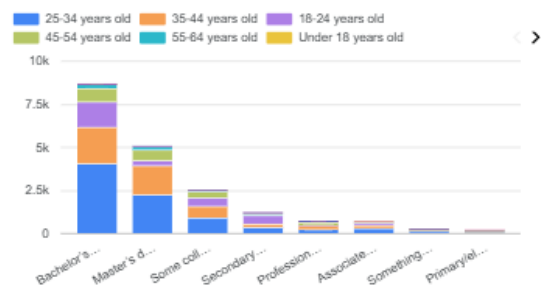
Record Count by Country



Record Count and Record Count by EdLevel



Record Count by EdLevel and Age



Top 10 Languages Current and Future – Python code below:

```
C: > Users > Brendan > OneDrive > Documents > IBM Data Analyst > Bar Charts Top 10 Languages.py > ...
1  import pandas as pd
2  import matplotlib.pyplot as plt
3
4  # Load your survey data
5  df = pd.read_csv(r"C:\Users\Brendan\Downloads\survey-data.csv")
6
7  # Split multiple languages into separate rows
8  lang_series = df['LanguageWantToWorkWith'].dropna().str.split(';').explode().str.strip()
9
10 # Count frequency of each language
11 lang_counts = lang_series.value_counts().head(10)
12
13 # Plot horizontal bar chart
14 plt.figure(figsize=(8,6))
15 plt.barh(lang_counts.index, lang_counts.values, color='skyblue')
16 plt.title("Top 10 Programming Languages (Want to Work With)")
17 plt.xlabel("Number of Respondents")
18 plt.ylabel("Language")
19 plt.gca().invert_yaxis() # largest at the top
20 plt.tight_layout()
21 plt.show()
```

Top 10 Databases to Work With Current and Future

```
C: > Users > Brendan > OneDrive > Documents > IBM Data Analyst > Top 10 Databases.py > ...
1  import pandas as pd
2  import matplotlib.pyplot as plt
3
4  # Load your survey data
5  df = pd.read_csv(r"C:\Users\Brendan\Downloads\survey-data.csv")
6
7  # Split multiple databases into separate rows
8  db_series = df['DatabaseWantToWorkWith'].dropna().str.split(';').explode().str.strip()
9
10 # Count frequency of each database
11 db_counts = db_series.value_counts().head(10)
12
13 # Plot horizontal bar chart
14 plt.figure(figsize=(8,6))
15 plt.barh(db_counts.index, db_counts.values, color='lightgreen')
16 plt.title("Top 10 Databases (Want to Work With)")
17 plt.xlabel("Number of Respondents")
18 plt.ylabel("Database")
19 plt.gca().invert_yaxis() # largest at the top
20 plt.tight_layout()
21 plt.show()
22
```

Job Postings

```

C: > Users > Brendan > OneDrive > Documents > IBM Data Analyst > Job Postings.py > ...
1  import pandas as pd
2  import matplotlib.pyplot as plt
3
4  # Load the Excel file
5  df = pd.read_excel(r"C:\Users\Brendan\OneDrive\Documents\IBM Data Analyst\job-postings.xlsx")
6
7  # Sort by number of jobs in descending order
8  df_sorted = df.sort_values(by="Number of Jobs", ascending=False)
9
10 # Plot horizontal bar chart
11 plt.figure(figsize=(10,6))
12 plt.barh(df_sorted["Location"], df_sorted["Number of Jobs"], color="steelblue")
13 plt.title("Job Postings by Location (Descending Order)")
14 plt.xlabel("Number of Job Postings")
15 plt.ylabel("Location")
16 plt.gca().invert_yaxis() # largest bar at the top
17 plt.tight_layout()
18 plt.show()
19
20
21

```

Popular Languages

```

C: > Users > Brendan > OneDrive > Documents > IBM Data Analyst > popular_languages.py > ...
1  import pandas as pd
2  import matplotlib.pyplot as plt
3
4  # Load the CSV file
5  df = pd.read_csv(r"C:\Users\Brendan\OneDrive\Documents\IBM Data Analyst\popular-language.csv")
6
7  # Inspect the columns to confirm structure
8  print("Columns in the file:", df.columns)
9
10 # Sort by salary in descending order
11 df_sorted = df.sort_values(by="Average Annual Salary", ascending=False)
12
13 # Plot horizontal bar chart
14 plt.figure(figsize=(10,6))
15 plt.barh(df_sorted["Language"], df_sorted["Average Annual Salary"], color="darkorange")
16 plt.title("Popular Programming Languages by Salary (Descending Order)")
17 plt.xlabel("Average Annual Salary")
18 plt.ylabel("Language")
19 plt.gca().invert_yaxis() # highest salary at the top
20 plt.tight_layout()
21 plt.show()
22

```